FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Chemistry — Class IX

Syllabus: Ch 1 — Nature of Chemistry | Ch 2 — Matter | Ch 3 — Atomic Structure | Ch 4 — Periodic Table & Periodicity | Ch 5 — Chemical Bonding | Ch 6 — Stoichiometry | Ch 7 — Electrochemistry | Ch 8 — Energetics | Ch 9 — Chemical Equilibrium

Total Marks: 65 **Time Allowed: 2 Hours 30 Minutes** **Version: 5**

General Instructions:

1. This paper has three sections: Section A (MCQs), Section B (Short Questions), and Section C (Long Questions).
2. Attempt Section A on the question paper by filling the bubble next to the correct option.
3. In Section B, attempt all 11 questions. For each, choose either the main question OR its alternative.
4. In Section C, attempt all 4 questions. For each, choose either the main question OR its alternative. Show all working; marks are awarded for method, not only the final answer.

Atomic Masses (amu): H=1, C=12, N=14, O=16, Na=23, Mg=24, Al=27, S=32, Cl=35.5, K=39, Ca=40, Mn=55, Fe=56, Cu=63.5 | Avogadro's Number $N_A = 6.022\times10^{23}$

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	Which branch of chemistry protects water poisoned by soil/dust using sedimentation, filtration, and disinfection?	Environmental chemistry	Organic chemistry	Inorganic chemistry	Geochemistry	Ⓐ Ⓑ Ⓒ Ⓓ
2	Which salt's solubility **DECREASES** as temperature rises from 20°C to 100°C?	NaCl	KCl	NH_4Cl	$Ca(OH)_2$	Ⓐ Ⓑ Ⓒ Ⓓ
3	In an isotope symbol, the number at the **upper-left** of the element symbol represents the:	Atomic number	Mass number	Number of electrons	Ionic charge	Ⓐ Ⓑ Ⓒ Ⓓ
4	Radioactive isotope Cobalt-60 is commonly used to:	Diagnose thyroid problems	Image the brain	Irradiate and shrink cancer tumours	Trace blood flow	Ⓐ Ⓑ Ⓒ Ⓓ
5	Which displacement reaction will occur (oxidizing power $F_2 > Cl_2 > Br_2 > I_2$)?	$Cl_2 + 2NaF \rightarrow 2NaCl + F_2$	$Br_2 + 2KI \rightarrow 2KBr + I_2$	$I_2 + 2KBr \rightarrow 2KI + Br_2$	$I_2 + 2NaCl \rightarrow 2NaI + Cl_2$	Ⓐ Ⓑ Ⓒ Ⓓ
6	Which statement about noble-gas electron configurations (Table 4.1) is correct?	All follow ns^1np^5	Noble gases are highly reactive	He ($1s^2$) is the exception; others follow ns^2np^6	Noble gases exist as diatomic molecules	Ⓐ Ⓑ Ⓒ Ⓓ
7	Using the melting points (H_2O 0°C, ethanol −117°C, NaCl 801°C, $MgCl_2$ 714°C), which is a **covalent** compound?	Sodium chloride	Magnesium chloride	Ethanol	Sodium fluoride	Ⓐ Ⓑ Ⓒ Ⓓ
8	Molar mass of urea, $CO(NH_2)_2$? (C=12, O=16, N=14, H=1)	60 g/mol	46 g/mol	44 g/mol	76 g/mol	Ⓐ Ⓑ Ⓒ Ⓓ
9	Na_2CX_3 has a formula mass of 106 amu (Na=23, C=12). Atomic mass of X is:	106	23	16	12	Ⓐ Ⓑ Ⓒ Ⓓ
10	Silver decorative objects tarnish (turn blackish) in air due to formation of:	Silver oxide	Silver sulphide	Silver chloride	Silver carbonate	Ⓐ Ⓑ Ⓒ Ⓓ
11	The branch of chemistry studying energy changes (heat evolved/absorbed) in reactions is called:	Thermodynamics	Chemical energetics	Electrochemistry	Photochemistry	Ⓐ Ⓑ Ⓒ Ⓓ
12	$N_2(g) + 2O_2(g) \rightleftharpoons 2NO_2(g)$ is produced industrially using:	An iron catalyst	A vanadium pentoxide catalyst	An electric spark, no catalyst	A platinum catalyst	Ⓐ Ⓑ Ⓒ Ⓓ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. Choose either the main question OR its alternative.

Part	Question	Marks	OR	Alternative Question	Marks
2(i)	Define medicinal chemistry and polymer chemistry. State one example application of each from your textbook (e.g. drug–target interaction; nylon/polyethylene/Teflon).	**3**	**OR**	What is titration, and which branch of chemistry is it an application of? Name TWO other daily-life applications of analytical chemistry from your textbook.	**3**
2(ii)	Differentiate between a concentrated solution and a dilute solution, using brine (a concentrated salt solution) as an example.	**3**	**OR**	Using the solubility curves for KNO_3 and NaCl, describe how the solubility of each changes as temperature rises from 0°C to 100°C. Which salt is more temperature-sensitive?	**3**
2(iii)	Define heavy water. State how its melting point (3.81°C vs 0°C) and density (1.1044 vs 0.997 g/cm³) compare with ordinary water.	**3**	**OR**	State the maximum electrons that s, p, and d sub-shells can each hold. Hence find the maximum electrons the M-shell (n=3) can accommodate.	**3**
2(iv)	An unknown element has valence configuration $3s^2 3p^4$. Determine its group and period, and state whether it is a metal or non-metal.	**3**	**OR**	An unknown element has valence configuration $3s^1$. Determine its group and period, and state whether it is a metal or non-metal.	**3**
2(v)	Name FOUR industrial catalysts from your textbook, along with the process or reaction each is used in.	**3**	**OR**	State how metallic character changes (a) down a group and (b) across a period, giving a reason for each trend.	**3**
2(vi)	Using lone-pair donation, explain how a coordinate covalent bond forms in the ammonia–boron trifluoride (NH_3–BF_3) adduct, given that BF_3 is electron-deficient.	**3**	**OR**	Using the melting points (water 0°C, ethanol −117°C vs NaCl 801°C, $MgCl_2$ 714°C), explain why covalent compounds have much lower melting points than ionic compounds.	**3**
2(vii)	State THREE industrial uses of graphite (other than pencils) and explain the structural property responsible for each.	**3**	**OR**	Silicon dioxide (quartz) has a giant covalent structure like diamond. Explain why this gives it high hardness and melting point, and name one industrial use.	**3**
2(viii)	Calculate the moles present in 31.6 g of potassium permanganate, $KMnO_4$. (K=39, Mn=55, O=16)	**3**	**OR**	Calculate the mass of 0.4 moles of ammonium chloride, NH_4Cl. (N=14, H=1, Cl=35.5)	**3**
2(ix)	State the oxidation states typically shown by Group 1, Group 2, and Group 13 elements (Table 7.2). What is the usual oxidation state of hydrogen, and when is this reversed?	**3**	**OR**	Identify the oxidizing and reducing agent in $Sn(s) + 2AgNO_3(aq) \rightarrow Sn(NO_3)_2(aq) + 2Ag(s)$, justifying with oxidation number changes.	**3**
2(x)	State any THREE Key Points about chemical energetics from your textbook's Key Points section.	**3**	**OR**	Define chemical energetics. Name TWO metals mentioned in the textbook's introduction whose large-scale production relies on huge amounts of released energy.	**3**
2(xi)	A hydrocarbon contains 80% carbon and 20% hydrogen by mass; molar mass 30 g/mol. Determine its empirical and molecular formula. (C=12, H=1)	**3**	**OR**	Balance the following equation: $C_2H_6(g) + O_2(g) \rightarrow CO_2(g) + H_2O(g)$	**3**

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	Alternative Question	Marks
3(i)	(a) Isotope A of an element has natural abundance 75.77% and isotopic mass 35 amu. If the element's relative atomic mass is 35.5 amu, find the isotopic mass of isotope B (abundance 24.23%). Show working. [3] (b) Describe the formation of anions from a sulphur atom (Z=16) and a chlorine atom (Z=17), showing electrons gained and the resulting charge. [2]	**5**	**OR**	(a) Element M has three isotopes: mass 24 (abundance 79%), mass 25 (abundance 10%), mass 26 (abundance 11%). Calculate the relative atomic mass of M. [3] (b) Compare protons, neutrons, and electrons passed through an electric field, stating the charge and relative mass of each. [2]	**5**
3(ii)	(a) A compound contains 85.7% carbon and 14.3% hydrogen by mass; molar mass 56 g/mol. Find its empirical and molecular formula. (C=12, H=1) [3] (b) Balance: $P_4(s) + O_2(g) \rightarrow P_2O_5(s)$ [2]	**5**	**OR**	(a) Transform into a net ionic equation, identifying spectator ions: $BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$ [3] (b) Calculate the mass of 0.3 moles of magnesium sulphate, $MgSO_4$. (Mg=24, S=32, O=16) [2]	**5**
3(iii)	(a) Determine the oxidation state of chromium in $Cr_2(SO_4)_3$ and of nitrogen in $NaNO_3$. Show working. [3] (b) Identify the oxidizing and reducing agent in $2Al(s) + 3CuSO_4(aq) \rightarrow Al_2(SO_4)_3(aq) + 3Cu(s)$, justifying with oxidation number changes. [2]	**5**	**OR**	(a) An unknown element has configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$. Find its group, period, block, and whether it is a metal or non-metal. [3] (b) State TWO differences between transition elements and s-block elements, with one example each. [2]	**5**
3(iv)	(a) In diamond each carbon bonds to four others in a rigid 3-D network (C–C length 1.544×10^{-10} m, angle 109.5°); in graphite each carbon bonds to only three others in 2-D layers held by weak forces. Explain why diamond is hard and insulating while graphite is soft, slippery and conducting. [3] (b) State TWO uses of diamond (other than jewellery) and TWO uses of graphite, linking each to a property. [2]	**5**	**OR**	(a) Describe the structure of buckminsterfullerene (C_{60}, "buckyballs"): number of carbon atoms, arrangement of hexagons/pentagons, and bonds per carbon atom. [3] (b) Using the data ($Ca(OH)_2$: 0.173 g/100g water at 20°C, decreasing to 0.066 g at 100°C), explain why this makes calcium hydroxide useful as an antacid/for chemical burns. [2]	**5**

Invigilator's Signature: ____________________ Candidate's Name / Roll No.: ____________________

Chapters Covered: Ch 1 (Nature of Chemistry) | Ch 2 (Matter) | Ch 3 (Atomic Structure) | Ch 4 (Periodic Table) | Ch 5 (Chemical Bonding) | Ch 6 (Stoichiometry) | Ch 7 (Electrochemistry) | Ch 8 (Energetics) | Ch 9 (Chemical Equilibrium)

— End of Question Paper —